

Muhammad Usman Afzal

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Profile

Electronics engineer working between embedded hardware and data. Hands-on with STM32 firmware, FPGA, lab instrumentation and Linux, and with the data side: pipelines, dashboards and anomaly detection for sensor data. Two peer-reviewed publications. Seeking a curricular internship and thesis in Italy. Working language: English.

Education

MSc, Electronic Engineering (Electronics for Intelligent Systems, Big-Data and IoT)

University of Bologna, Italy | in progress

MS, Electrical Engineering

Information Technology University (ITU), Lahore, Pakistan | completed

BSc, Electrical Engineering

Information Technology University (ITU), Lahore, Pakistan | 2018 - 2022

- Dean's Honor List (Aug 2022); merit-based scholarship.

Experience

Research Associate, Micro Nano Lab, ITU Lahore

Aug 2022 - Sep 2024

- Led a student team on radiation profiling of Reconfigurable Intelligent Surfaces (RIS) to generate localization datasets.
- Built a low-cost, safe Electrical Impedance Tomography (EIT) device to produce data for biomedical machine learning.
- Implemented a low-cost RF testbed; worked on the design of a large-scale meta lens.

Teaching Assistant, ITU Lahore

Spring and Fall 2024

- Digital Logic Design and Applied Physics: supported students, held office hours, graded coursework and exams.

Onsite Trainee Engineer, RADA Technologies, Pakistan

Dec 2020 - May 2021

- Troubleshoot VFD and PLC systems and worked on CNC modernization; evaluated STM32 (Blue Pill / Black Pill) platforms.
- Built Raspberry Pi / Linux automation systems and Python machine-vision prototypes; helped set up lab infrastructure.

Projects

Edge anomaly detection on STM32 (in progress)

STM32, C, TinyML

- Vibration and motion sensor node that detects anomalies on the microcontroller. github.com/usmanafzal01

Sensor data pipeline with Grafana (in progress)

Python, time-series storage, Grafana

- Ingestion, storage and dashboards for the edge node's readings and anomaly flags.

Antenna radiation mapper (final-year project)

SDR, NI myRIO, LabVIEW

- Measured antenna radiation patterns with an SDR and myRIO; funded by ITU's Undergraduate Research Outreach Program.

Other builds

- FFT on FPGA with mean, variance and standard deviation; horn antennas for 2-3, 3-5 and 1-18 GHz bands; 2.8 GHz RF resonator.
- Six-floor lift controller (ATmega328P), PLC irrigation control, LabVIEW inverted pendulum, stepper control on NI myRIO.

Publications

- *Privacy and Security in Distributed Learning: A Review of Challenges, Solutions, and Open Research Issues*, IEEE Access, 2023.

- *Energy Efficient and Secure Federated Learning for UAV Swarms*, 7th IEEE ICECE, 2024 (paper presentation and poster competition).

Skills

- **Embedded:** C, C++, STM32 (Keil, STM32 IDE), Arduino, RISC-V basics, Raspberry Pi, Linux, perf profiling.
- **Digital and data:** Verilog/VHDL, Python, MATLAB, PyTorch, TensorFlow, signal processing, Grafana.
- **Hardware and lab:** Altium, NI Multisim, SolidWorks, AutoCAD, LabVIEW, PLC; oscilloscopes and bench instruments; CST, HFSS, COMSOL.

Certifications, affiliations and awards

- DeepLearning.AI: Convolutional Neural Networks; Improving Deep Neural Networks; Structuring Machine Learning Projects; Machine Learning in Production.
- NASA International Space Apps Challenge 2025: local finalist (team Debugging Masters). Member of IEEE and Pakistan Engineering Council.